

Sea Scallop Life History Parameters

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1 Sea Scallop Growth

This is an update on the growth estimates of Hart & Chute (2009a), and follows the methods there except as specifically noted. In brief, the data consist of the shell heights of successive growth rings on the upper valve of sea scallop shells. Because early growth rings may be obscure or missing, these data are used here as a measure of annual growth increments, rather than an estimate of absolute age. The growth rings have been verified to be annual (Hart & Chute 2009b, Chute et al. 2012).

A linear mixed-effects model (using lme4 in *R*) that predicts the shell height of the larger ring of an increment r_2 from the starting ring shell height r_1 is used to estimate the von Bertalanffy coefficients L_∞ and K . Random effects on both the slope and intercept, on the level of individual scallop, take into account individual variability in both K and L_∞ , which otherwise would cause bias and artificially inflate sample size. Estimates of individual variation from the random effects are used, together with the estimates of the means of these parameters, to construct stochastic growth matrices for the CASA, SAMS and SYM models. The von Bertalanffy parameter t_0 is not estimatable by these methods, but it is not required for a size-structured assessment.

A station-level random effect on the intercept only was included so that there are nested station and individual scallop random effects. The term accounts for the fact that scallops within a station are more likely to have similar growth. Including the station-level effect mainly affected the estimated variability in L_∞ ; not including it (as was done in Hart & Chute 2009a), biases this variability slightly low. In addition, not including the station effect biases the standard errors low since sample size is somewhat inflated. However, the station-level random effect did not measurably affect the estimates of the mean L_∞ and K parameters.

The shells used for this study were collected between 1982 and 2023. However, most were collected after 2000, so growth estimates prior to that time will carry considerable uncertainty. Increments with a starting shell height less than 40 mm, and those from Canada and the slow growing Nantucket Lightship South scallops were excluded from the analysis. The Mid-Atlantic dataset consisted of 21583 growth increments from 6600 shells and 1392 stations, and the Georges Bank set had 56561 growth increments from 12566 shells and 2378 stations. There was good spatial coverage (Figure LH-2).

Mixed-effects GAM models (using the *R* package gamm4) were employed to explore variability in growth by year, applied to the Mid-Atlantic, and Georges Bank Open and Closed:

$$r_2 \sim r_1 + s(D, L) + s(y) \quad (\text{LH-1})$$

where r_1 and r_2 are the starting and final shell heights of an increment, D , L and y are depth, latitude, and the year in which the 2nd ring of an increment is laid down, respectively. These models were used to predict one year's growth of a scallop at an intermediate shell height (Figure LH-3). Growth in the Mid-Atlantic was the slowest in the early 1990s and then tended to increase afterwards, whereas growth has been slowing in recent years on Georges Bank. These results were used to partition the time series into periods of similar growth. For the Mid-Atlantic, four time blocks were chosen, and two each for the Georges Bank Closed and Open areas (Table LH-1). For each region, period was included as a factor in the basic mixed-effects regression:

$$r_2 = a_0 + b_0 r_1 + a_p \text{as.factor}(\text{period}) \quad (\text{LH-2})$$

This formulation assumes that it is L_∞ which varies among periods. Results are shown in Table LH-1 and Figure LH-3.

2 Shell height to meat weight relationships

This section is an update of Hennen & Hart (2012). Meat weight data is obtained from two sources: surveys and at-sea observers. The survey data will be analyzed first.

Shell height to meat weight relationships from survey data

Since 2001, a sample of scallops from the catch (typically 3-10 scallops on about half the stations) on the NEFSC and VIMS dredge surveys are dissected, and their shell height, whole weight, gonad weight and meat weights are measured; in most cases, these were the same scallops that were analyzed for growth.

Meat weights vary seasonally, and the survey timing has varied. Prior to 2009, the Mid-Atlantic survey was conducted in late June and/or July, but since then, except 2020, the survey has been conducted in May and/or early June (Figure LH-5). The Georges Bank component was conducted in July to early August, but now is typically conducted in June.

Mixed-effects GAM models were used to account for survey timing, both without and with spatial covariates:

$$W = \ln(H) + s(J) \quad (\text{LH-3})$$

$$W = \ln(H) * D + s(J) + \text{as.factor}(Y) + L \quad (\text{LH-4})$$

where W is meat weight, H is shell height, J is Julian day, Y is year, and L is latitude, using a gamma distribution with a log link; there were also station-level random effects on the intercept only.

The seasonal cycle of meat weights is different in the two regions (Figure LH-6). Mid-Atlantic scallops typically have a spring spawn in April or early May, and hence meats

weights are low on May 1. The meat weights have been lower in the spring in recent years. The meats appear to peak around the end of June. On Georges Bank, meat weights are fairly stable until dropping in August.

For stock assessment modeling purposes, meat weights were predicted using 2010 as the baseline year on Julian day 150 in the Mid-Atlantic, and day 180 for Georges Bank. Comparisons of these curves with those used in previous assessments are shown in Figure LH-7.

Commercial shell height/meat weight data

Shell height to meat weight relationships on commercial vessels can differ from the survey relationships for two primary reasons. First, as discussed above, meat weights vary seasonally, and the fishery is conducted year-round, not just when the survey occurs. Secondly, scientist on the survey carefully shuck the scallops, taking all the meat. Scallopers shuck for speed, often leaving a small portion of the meat in the shell.

For these reasons, commercial meat weights are monitored by fishery observers. Once a watch, the observers take about 100 scallops retained for landing, and measure their shell heights. The scallops are then given to a fisherman to shuck, and the scallop meats are put into a graduated cylinder to get a volumetric measure. The volume is converted to a weight using a specific weight of 1.05 (Caddy & Radley-Walters 1972, Smolowitz et al. 1989). Only samples with between 75 and 125 meats were used. Additionally, a few outliers, where the meat weight anomalies (see below) were less than -0.67 or greater than 0.67 were removed. The analysis below is based on the cleaned data set, which comprises about 95% of the original data.

The shell height measurements are used to compute the predicted meat weight, using the estimates from survey data with covariates (depth and latitude), but estimated at fixed Julian days (150 for the Mid-Atlantic and 180 for Georges Bank) and year (2010). The data are then aggregated by month and region and “clop” (Georges Bank and Mid-Atlantic, open or rotational access area), to compute an empirical monthly meat weight anomaly A_m (Hennen and Hart 2012).

$$A = \frac{\text{predicted} - \text{observed}}{\text{predicted}}. \quad (\text{LH-5})$$

A generalized additive model (GAM) was developed for predicting the meat weight anomaly:

$$\text{MWanomaly} \sim s(\text{Year}, \text{Month}) + \text{nscallop} + \text{Clop}, \quad (\text{LH-6})$$

where s is a two dimensional smoother on Year and Month, nscallop is the number of meats in the observer’s sample, and Clop is 1 if the sample was taken in a rotational access area, and 0 if it was taken in an open area. The term “nscallop” was used since there is a slight tendency for the anomalies to decline with the number of scallops in the sample. This might occur if the probability that some of the measured scallops didn’t get into the graduated cylinder increases with number of scallops measured. For the purposes of the predictions given below, I used nscallop = 100, which is both what is specified in the protocol, and also the most common number measured. Using the two dimensional smoother rather than separate smoothers on Year and Month substantially improved the fit and reduced AIC. This

indicates that there was a significant Year/Month interaction, i.e., the annual meat weight cycle significantly varied by year (Figure LH-8).

For the CASA assessment modeling, meat weight anomalies were computed for each year by taking the landings weighted mean anomaly for that year. For years prior to 1994, when there was no observer meat weight data, the mean meat weight anomaly by month was used, except for 1989 – 1993, for which used the mean anomaly from 1994 – 1999.

3 Natural Mortality

Estimates of natural mortality M can be made based on longevity, i.e., the age of the oldest observed animal $t_{\max}$. Here we will apply three such estimators:

$$M = \exp(1.23 - 0.834 \ln t_{\max}) \quad \text{Hoenig (1983)}$$

$$M = 4.899t_{\max}^{-0.916} \quad \text{Then et al. (2015)}$$

$$M = 5.4/t_{\max} \quad \text{Hamel \& Cope (2022)}$$

The Hoenig (1983) was based on data from mollusks, whereas the other two were based on fish. Assuming that the first visible ring is age 2, which is correct for most but not all shells, the oldest aged scallop in the Mid-Atlantic was 14 years old. Based on the three estimators, this gives estimates of $M = 0.38$, 0.44 and 0.39 , respectively. The working group concluded that $M = 0.4$ is a reasonable estimate for the Mid-Atlantic. This is considerably higher than the $M = 0.25$ estimate used in recent past assessments.

The oldest shell aged from Georges Bank was 19 years old, which gives M estimates of 0.29 , 0.33 , and 0.28 , respectively. The Georges Bank Closed CASA model, which because of low or zero fishing mortality in many years, has the ability to make a fairly precise estimate of M (Hart & Chang 2022), estimated $M = 0.27$. Because this estimate is comparable to those from the three above estimators, we used $M = 0.27$ in the Georges Bank Open CASA model and Georges Bank SYM model. This is higher than the estimate of $M = 0.2$ in previous recent assessments.

References

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Table and Figures

Region	Period	L_{∞}	K	SDL_{∞}	SDK	SEL_{∞}	SEK
MA	1990-99	125.3	0.570	7.1	0.13	0.35	0.003
MA	1975-1978; 1984-89; 2000-03	130.3	0.570	7.5	0.13	0.36	0.003
MA	1979; 1982-3; 2004-08; 2019-20	134.3	0.570	7.9	0.13	0.37	0.003
MA	2009-12; 2021-22	136.9	0.570	8.2	0.13	0.37	0.003
GBCI	1986-97; 2015-22	142.7	0.444	9.6	0.12	0.34	0.12
GBCI	1975-85; 1998-2014	147.5	0.444	10.3	0.12	0.35	0.12
GBOp	1987-90; 2011-22	140.7	0.455	8.1	0.11	0.16	0.002
GBOp	1975-86; 1991-2010	143.4	0.455	8.5	0.11	0.17	0.002

Table LH-1. Growth parameters by period and region.

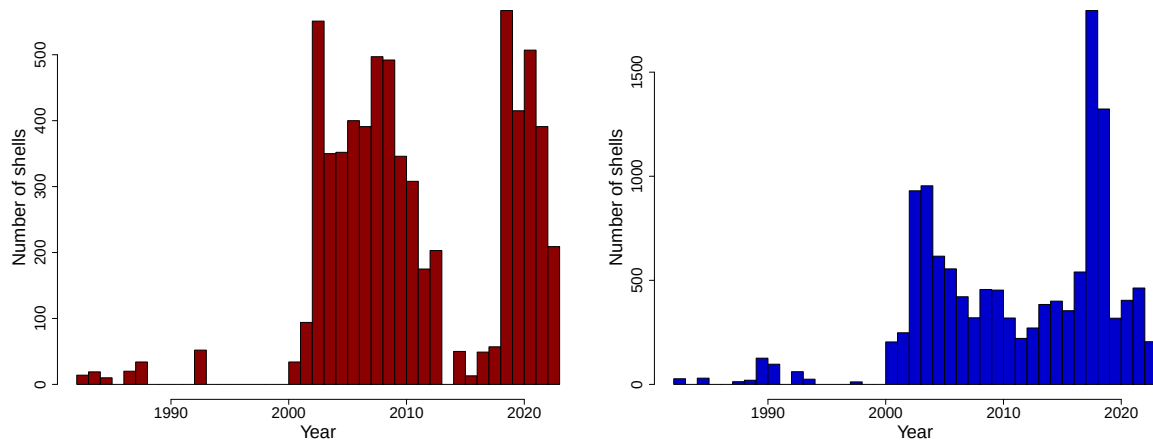


Figure LH-1. Number of shells analyzed by year for the Mid-Atlantic (left) and Georges Bank (right)

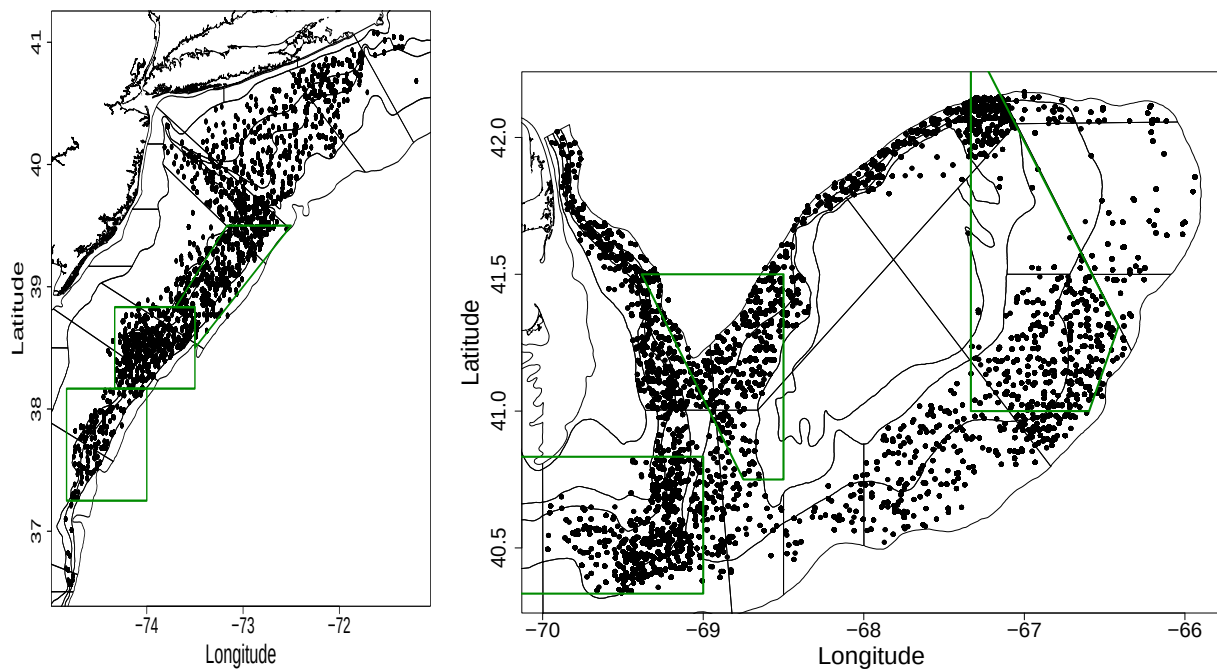


Figure LH-2. Spatial location of shell samples for the Mid-Atlantic (left) and Georges Bank (right)

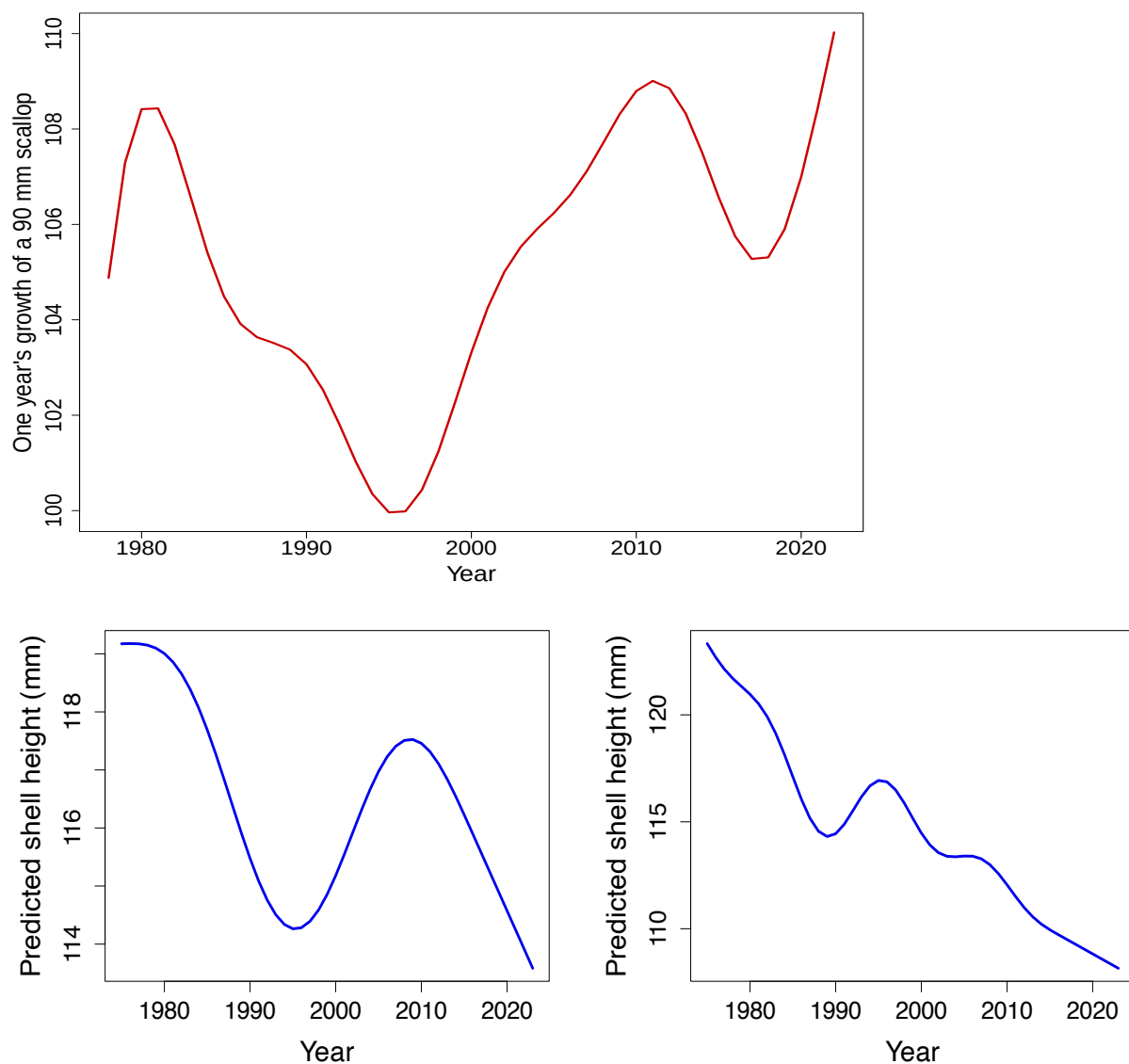


Figure LH-3. GAM predicted one year's growth of a 90 mm scallop (Mid-Atlantic) and a 100 mm scallop (Georges Bank) by year for the Mid-Atlantic (above), Georges Bank closed (lower left) and Georges Bank open (lower right)

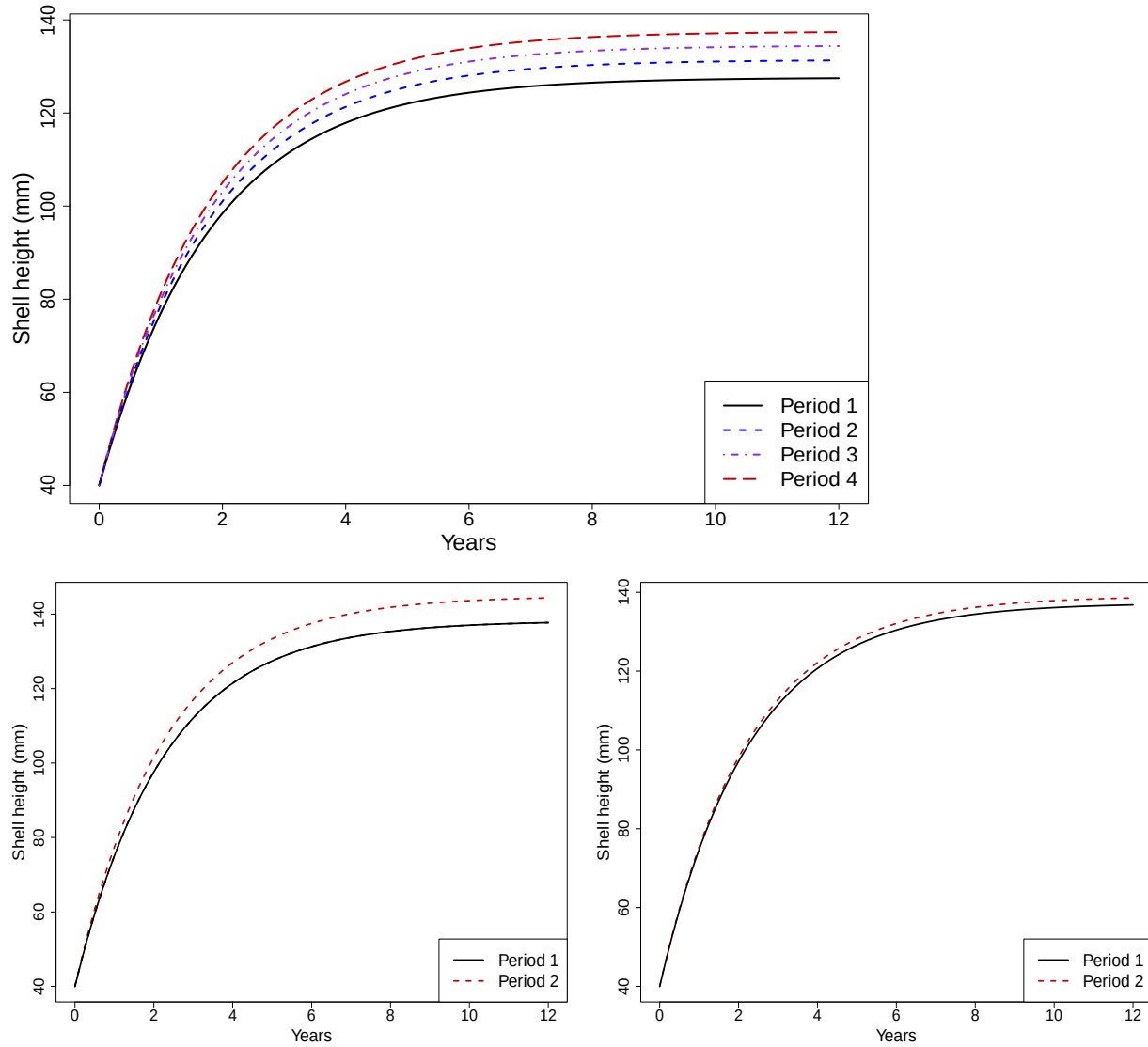


Figure LH-4. GAM predicted one year's growth of a 90 mm scallop (Mid-Atlantic) and a 100 mm scallop (Georges Bank) by year for the Mid-Atlantic (above), Georges Bank closed (lower left) and Georges Bank open (lower right)

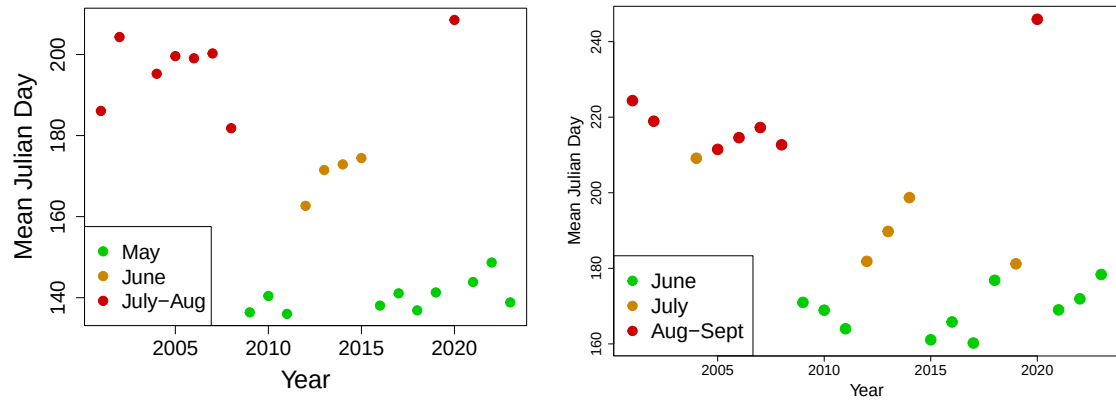


Figure LH-5. Meat weight sample timing by year for Mid-Atlantic (left) and Georges Bank (right).

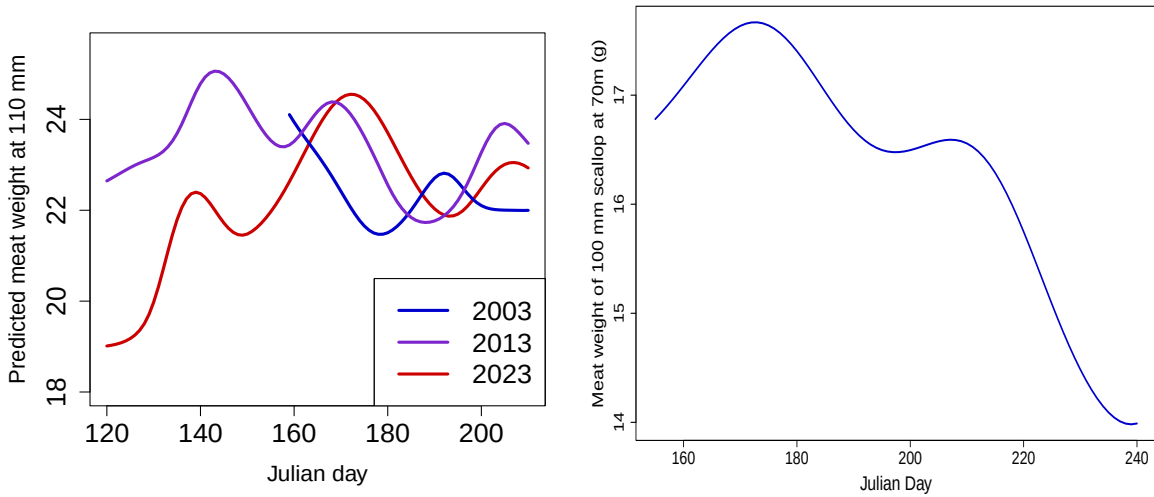


Figure LH-6. Seasonal effects on meat weight during the survey period for Mid-Atlantic (left) and Georges Bank (right)

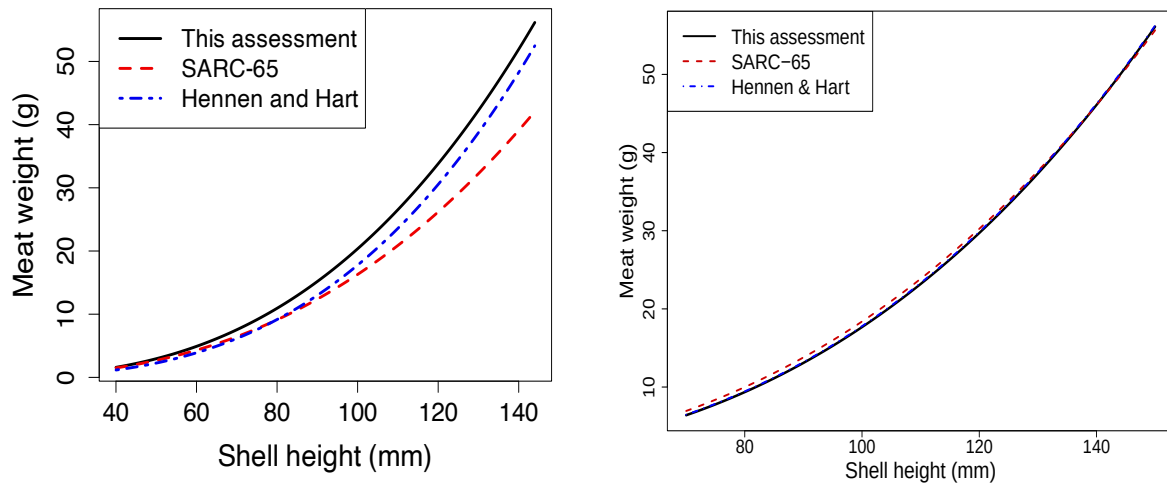


Figure LH-7. Comparison of the current shell height relationships with those from SARC-65 and Hennen and Hart (2012)/SARC-50 for the Mid-Atlantic (left) and Georges Bank (right)

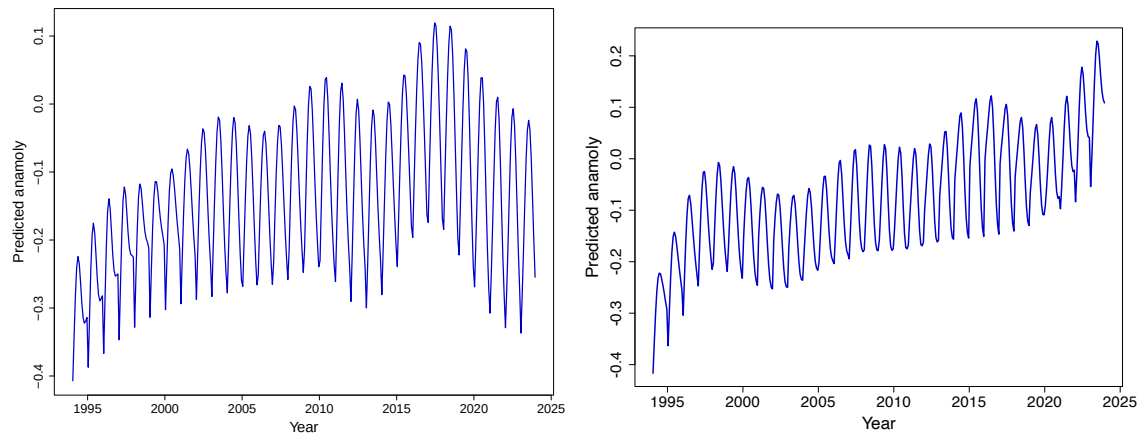


Figure LH-8. Estimated meat weight anomaly by year and month for Mid-Atlantic (left) and Georges Bank (right)